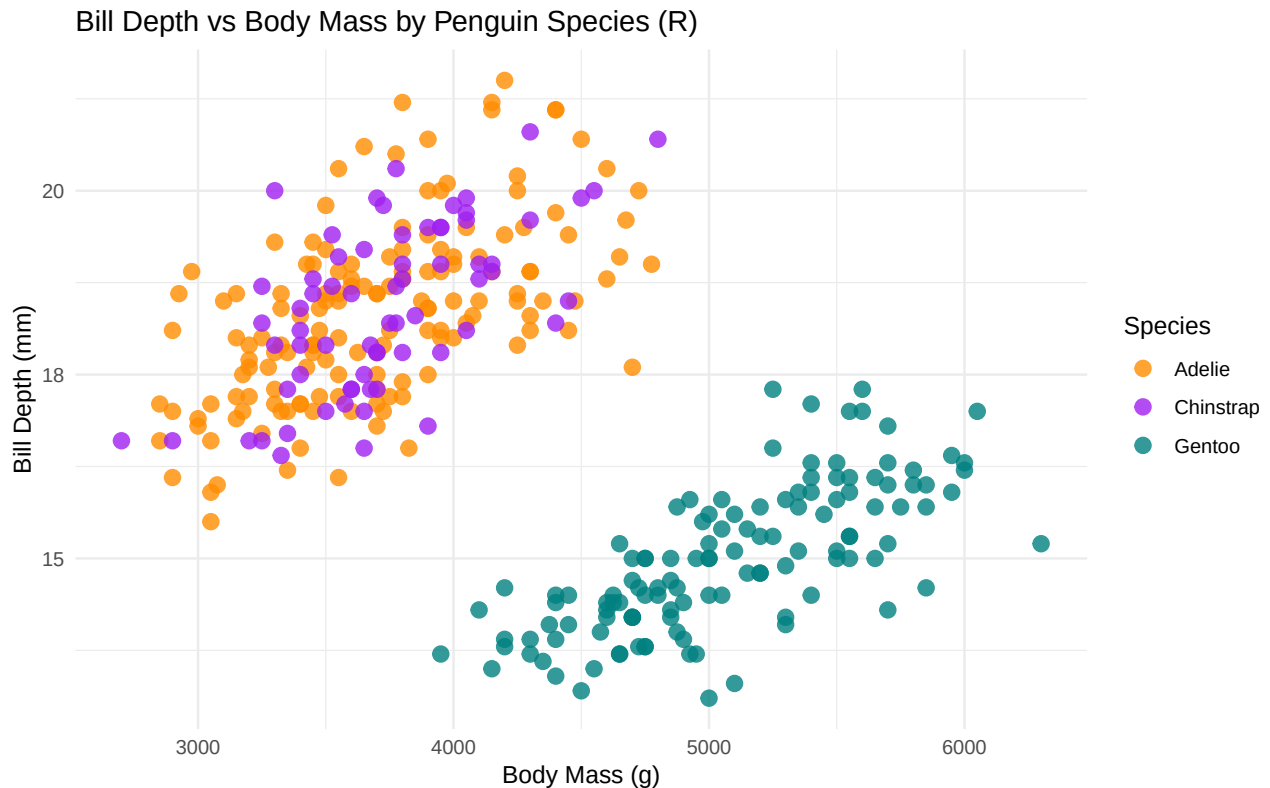


Palmer Penguins Analysis - Bill Depth vs Body Mass

2025-03-12

R Visualization

```
# Create scatter plot with R
ggplot(penguins_clean, aes(x = body_mass_g, y = bill_depth_mm, color = species)) +
  geom_point(alpha = 0.8, size = 3) +
  scale_color_manual(values = c("Adelie" = "darkorange", "Chinstrap" = "purple", "Gentoo" = "#008080"))
labs(
  title = "Bill Depth vs Body Mass by Penguin Species (R)",
  x = "Body Mass (g)",
  y = "Bill Depth (mm)",
  color = "Species"
) +
theme_minimal()
```



Python Visualization

```
# Import libraries
import pandas as pd
import numpy as np
```

```

import matplotlib.pyplot as plt

# Access the penguins data from R
penguins = r.py_penguins

# Define colors for each species
colors = {"Adelie": "darkorange", "Chinstrap": "purple", "Gentoo": "#008080"}

# Create figure and axis
fig, ax = plt.subplots(figsize=(8, 5))

# Create scatter plot for each species using only matplotlib
for species in penguins['species'].unique():
    # Filter data for this species
    species_data = penguins[penguins['species'] == species]

    # Plot this species
    ax.scatter(
        species_data['body_mass_g'],
        species_data['bill_depth_mm'],
        c=colors[species],
        label=species,
        alpha=0.8,
        s=80
    )

# Add title and labels
ax.set_title("Bill Depth vs Body Mass by Penguin Species (Matplotlib)")
ax.set_xlabel("Body Mass (g)")
ax.set_ylabel("Bill Depth (mm)")

# Add legend
ax.legend(title="Species")

# Add grid for better readability
ax.grid(True, linestyle='--', alpha=0.7)

# Adjust layout and show
plt.tight_layout()
plt.show()

```

